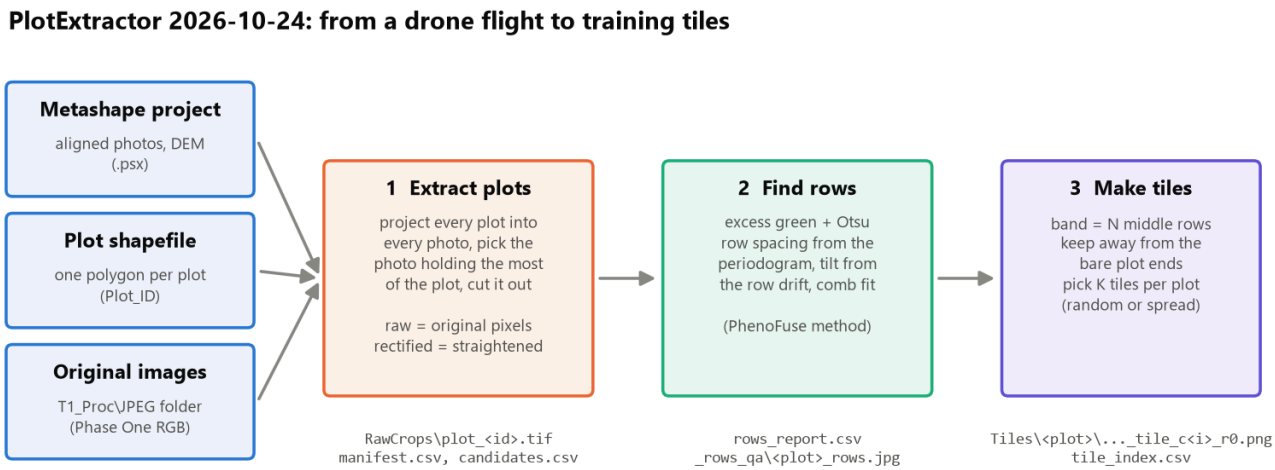
PlotExtractor 2026-10-24

Technical report

Middle-row training tiles with PhenoFuse row detection, original-quality plot extraction from Phase One images (JPEG and IIQ-Direct), and the experiments on the York 2025 calibration trial.



PlotExtractor 2026-10-24 · DPIRD / APPN · 24 October 2026

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Summary

- **New:** middle-N-rows tiling on the Training tiles tab, using the PhenoFuse RGB row detection adapted to raw photos (tilt-aware), with tiles per plot (random or spread) and an end distance that keeps tiles off the bare plot ends.
- **Original quality:** raw masked crops of all 36 York plots are identical to the original T1_Proc JPEGs (686,638,625 pixels, max difference 0). Rectified crops are resampled (~5% lower resolution, colour unchanged).
- **Rows:** 5 sown rows found in all 36 plots in both the raw and rectified sets; spacing 204-220 mm (mean 212.6 mm); middle rows 2-4 of 5 everywhere.
- **Tile bias fixed:** tiles now sit inside the usable length (mean 3.5 m of a ~4.5 m plot); 288 tiles re-checked: none outside the usable length, all pixel-identical to the crop.
- **Coverage:** 4 of 36 plots are never whole in a single photo (0.4-3.9% missing) because the photo's short side ran along the plots; turning the camera 90° fixes this for future flights.
- **Also fixed:** the margin now widens rectified crops; masked crops keep plot + margin; the manifest records the source pixel box of every raw crop.
- **IIQ-Direct (added 26 Oct 2026):** plot crops developed straight from the T0_Raw IIQ files with the Phase One ImageSDK and APPN's HR settings. On the 29 York plots cut from the same photo they are aligned with the JPEG crops to 0.02 px, have ~12% less noise, ~23% more fine detail (detail/noise 7.4 vs 5.4) and a slightly different colour rendering (less green: green cover 2.3% vs 3.0%, same plot ranking, r = 0.996). Highest image quality; do not mix the two sources in one analysis (E7).

1. Aims

1. Add an option to the Training tiles tab that tiles only the N middle rows of each plot (e.g. 3 of 5 or 7), with tiles selected at random within those rows, using the row detection of PhenoFuse 4.0.2.
2. Check that plot-wise raw images are 100% the same quality as the original images.
3. Extract raw and rectified plot images of the York calibration trial and make middle-3-row tiles from both.
4. (Follow-up) Keep tiles away from the bare plot ends, let the user choose how many tiles per plot, and explain why some plots are not whole in one photo.
5. (26 Oct 2026) Compare the plot images cut from the T1_Proc JPEGs with plot images developed directly from the T0_Raw IIQ files (IIQ-Direct, APPN HR settings), and check the quality of both.

2. Data

Item	Value
Trial	2025 Calibration trial I SIF DPIRD, York (wheat block)
Flight	Phase One, 25 June 2025, run00; 763 aligned photos; ~13 m above ground
Original images	G:\APPN 2025\...\2025-06-25\run00\T1_Proc\JPEG (8-bit RGB, Adobe RGB profile)
Raw IIQ files	G:\APPN 2025\...\2025-06-25\run00\T0_Raw (764 IIQ, Phase One iXM-GS120, 12768 x 9564; only read)
Metashape project	T1_Proc\MetaShapeProcessing.psx (DEM present)
Plots	edited_grid_edited_edited_edited_edited.shp: 36 plots (6 ranges x 6), ~1.1 x 4.5 m, 5 sown rows
Note on the given shapefile	The 'Calibtration trial shape file' holds only the trial boundary (one polygon, ~21.5 x 77 m). An even split did not match the field, and PhenoFuse grid_auto found no vegetated bank at this seedling stage, so the edited 36-plot wheat grid was used; it matches the orthomosaic.
Extraction	margin 0.10 m, GeoTIFF (LZW), no sRGB conversion, 8 workers; raw = Masked to plot (14 s), rectified = Rectified rectangle (36 s)
Output	F:\Ibrahim's Workspace2\Plot extraction processing\Calibration trail 2025\PlotExtractor_2026-10-24_York_2025-06-25

3. Methods

3.1 Plot extraction

Each plot polygon (plus margin) is dropped onto the DEM and projected into every aligned camera with the full camera model. Each camera is scored by the share of the plot it images, and the plot is cut from the best one. Masked, Native and Bounding-box styles copy a pixel box of the photo unchanged (Masked sets pixels outside plot + margin to 0). Rectified rectangle renders the plot on an upright ground grid through a projected mesh (96 px cells, bicubic), filling what the first photo misses from the next ones with a minimum-error seam.

Changes in this version: the manifest records raw_style and the source pixel box src_x0, src_y0, src_x1, src_y1 of every raw crop (crop pixel (0,0) = photo pixel (src_x0, src_y0)); rectified crops use the margin-expanded ring; masked crops mask to plot + margin instead of the bare plot.

3.2 Row detection (row_detect.py)

Ported from PhenoFuse 4.0 phenoapp/core/grid_rows.py (_row_signal, row_pitch_from_profile, rows_from_profile), pure numpy so it fits the PyInstaller build without scipy. Steps:

1. Reduced copy of the crop (~1200 px across). ExG = (2G−R−B)/(R+G+B); vegetation = ExG above one Otsu threshold per plot. Profiles across the plot: share of vegetation pixels and mean ExG, each detrended to a ridge signal.
2. Pitch: the plot is split into K = clip(length/250, 6, 24) segments along its length; the power spectra of the segment signals are summed (nfft 65536) and the peak inside 170-320 mm is taken. Summing powers removes the effect of tilt on the pitch.
3. Tilt: the row phase of each segment, (−arg(z)/2π)·P mod P, is unwrapped along the plot and fitted with a weighted least-squares line (outliers dropped over 4 iterations); tilt = atan(slope), limited to ±8°.
4. Fine pitch: on the de-tilted whole-plot profile, the pitch is searched over P·[0.94, 1.06] (241 steps) for the largest coherent power. This removes the ~4.5% FFT-bin quantisation.
5. Comb: teeth at that pitch fill the span; the phase is fitted; every tooth gets a contrast (ridge at the tooth minus the mid-gaps). Teeth outside the image data are never rows.
6. Plot rows: with sown rows given, the best-contrast block of that many teeth fully inside the data; otherwise the run of present teeth excluding teeth within 0.3 pitch of the edge.
7. middle_band: the block of N rows closest to the centre of the plot's rows; band = first row − P/2 to last row + P/2.

3.3 Tile placement (tiler.py)

1. Across: square tiles with side = band width, following the measured tilt (the band centre moves by tan(angle) per pixel along the plot).
2. Along: data ends = first/last position where the band is fully imaged; plot ends = data ends moved in by the extraction margin (manifest margin_m); planted ends = where the smoothed vegetation share in the band reaches 35% of its median over the central half (used only if the planted length is at least half the plot); usable = the tighter ends moved in by the end distance (default 0.5 m).
3. n = floor(usable / side) tiles, centred in the usable length; then K tiles per plot are picked (random with a seed, or evenly spread).
4. Tiles are read straight from the crop and saved losslessly; rows_report.csv and a QA picture are written per plot.

3.4 Fidelity check (fidelity_check.py)

Raw: each crop is compared pixel for pixel with the box of its source JPEG recorded in the manifest, decoded with the same decoder (Pillow/libjpeg-turbo); pixels inside plot + margin must be identical; the ICC profile is compared byte for byte. Rectified: the source box is registered to the crop (ORB + RANSAC homography), and local patches are aligned with sub-pixel phase correlation; the scale (source px per output px) and the mean RGB shift against the raw crop are reported.

3.5 Frame-coverage analysis (why_cut.py)

Every plot (+0.10 m margin) was projected into all 763 photos with the Metashape camera model; for each plot the photo with the smallest overshoot past the frame edge was kept, with the plot's span and offset in that photo. Camera centres were grouped into flight lines to get the shot spacing and line spacing; ground heights came from the DEM.

3.6 IIQ-Direct extraction with the APPN HR settings

An IIQ file is the camera's raw sensor record: one colour per pixel behind a Bayer filter, 14-bit, no colour rendering yet. The T1_Proc JPEGs were made from these files once by Phase One's software, as 8-bit, lossy-compressed, Adobe RGB images. IIQ-Direct skips that JPEG: for each plot it develops only the plot's pixel box straight from the IIQ, with the Phase One ImageSDK (4.1.84), and saves the result losslessly. The method follows APPN-UQ's PhaseOne_Pipeline (03_backprojection.py + HR_clip.exe, MIT licence, appn@uq.edu.au); PlotExtractor's p1tool is adapted from their HR_clip code and credited in the app, the README and THIRD_PARTY_NOTICES.md.

Where the conversion happens. Both methods find the plot the same way (same Metashape alignment, same projection, same photo choice). They differ only in where the raw sensor data is turned into a colour picture ("development"). In our JPEG-based method this happens once per photo, for the whole frame, in Phase One's software before PlotExtractor runs; PlotExtractor only cuts the finished JPEG (Figure A). In IIQ-Direct the JPEGs serve only to align the photos; the plot's pixels are developed inside PlotExtractor, from the IIQ, for the plot's box only (Figure B). APPN's own pipeline converts twice: IIQ to 0.3-scale TIFFs (no noise reduction or sharpening) for alignment, then HR_clip for the plot box; IIQ-Direct reuses the second step and uses whatever images the project was aligned on.

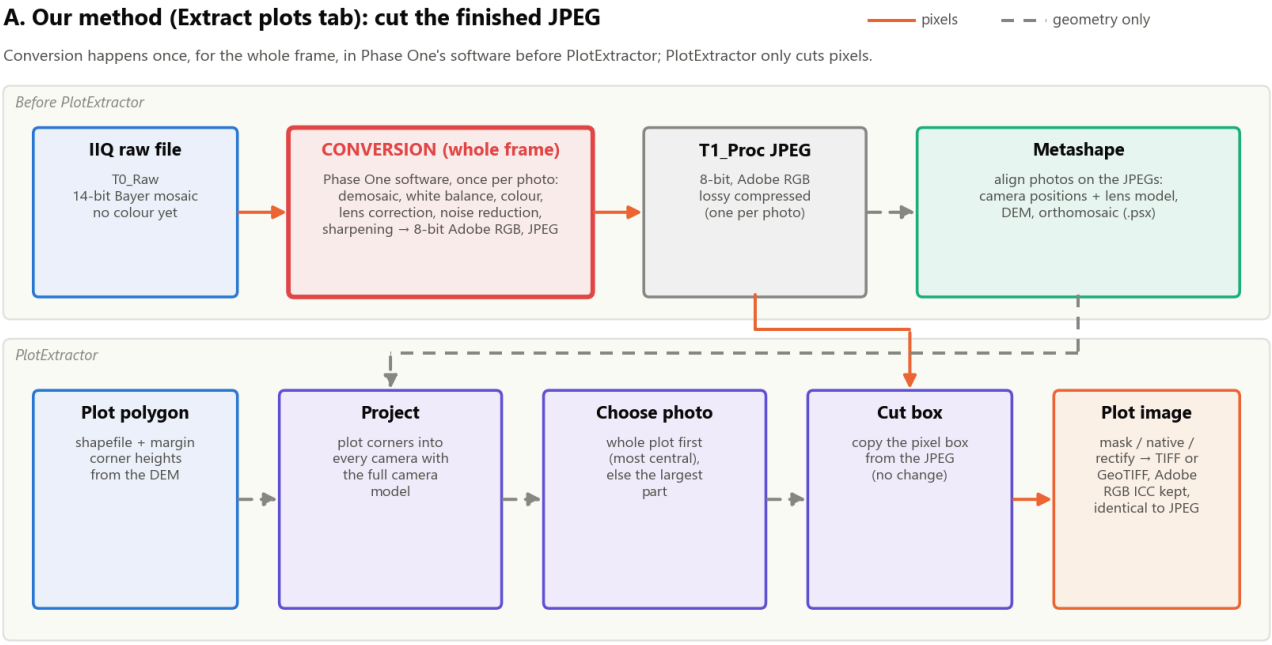


Figure 1. *A. Our method (Extract plots tab): the conversion is done before PlotExtractor; the plot is cut from the T1_Proc JPEG unchanged.*

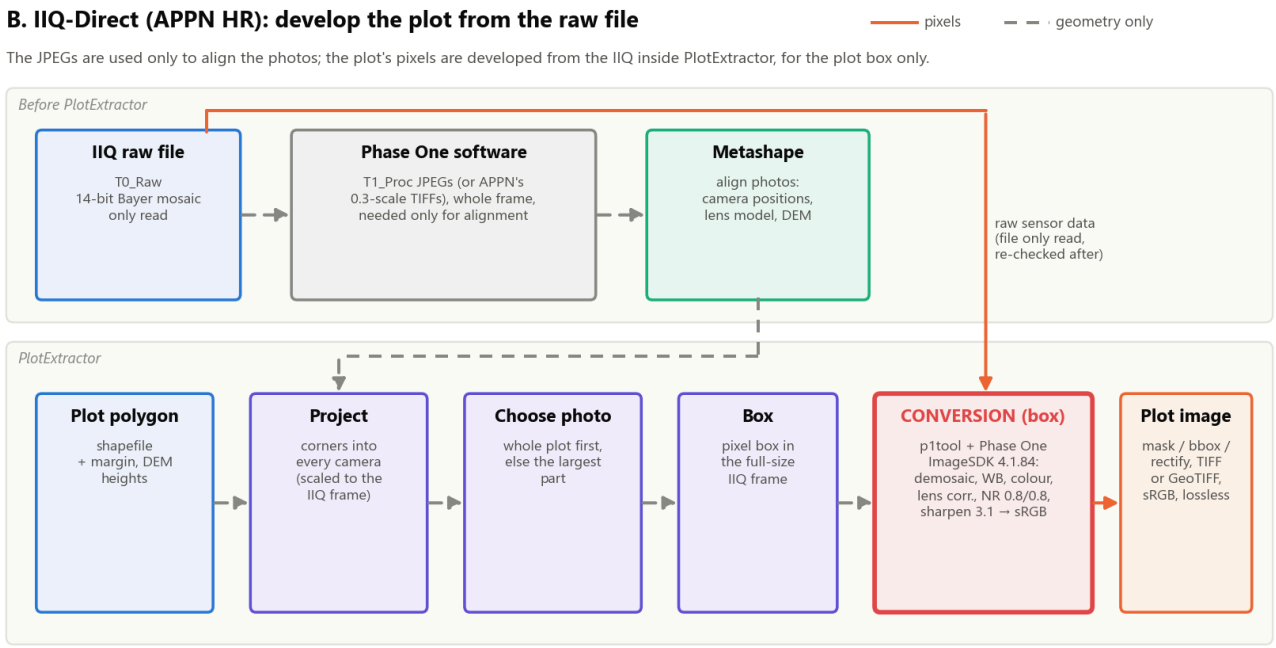


Figure 2. *B. IIQ-Direct (IIQ plot extraction tab, APPN HR): the JPEGs give only the geometry; the plot box is developed from the IIQ by the Phase One ImageSDK.*

Step	Our method (JPEG)	IIQ-Direct (APPN HR)
1. Raw capture	IIQ, 14-bit Bayer	same
2. Conversion	Phase One software, whole frame, before the trial is processed	p1tool + SDK, plot box only, during extraction
3. Alignment	Metashape on the JPEGs	same project (geometry only)
4. Find the plot	corners projected into every camera, photo chosen	identical
5. Get the pixels	copy the box from the JPEG	develop the box from the IIQ
6. Output	8-bit, Adobe RGB, has been through JPEG compression once	8-bit sRGB, never lossy-compressed

Pros and cons

	Our method (JPEG)	IIQ-Direct (APPN HR)
Pros	Fast (a cut: ~14 s for 36 plots). No extra software or SDK licence. Pixels are exactly the delivered T1_Proc images, consistent with the orthomosaic and earlier seasons. Bit-repeatable; QA proves diff 0.	Highest image quality: ~12% less noise, ~23% more fine detail, no lossy step (E7). Development is a choice (HR for looks, Neutral for least-processed pixels). Only the plot box is developed. Same geometry and QA as the JPEG method; IIQs untouched.

	Our method (JPEG)	IIQ-Direct (APPN HR)
Cons	Quality capped by the JPEG (8-bit, lossy; Phase One's NR and sharpening fixed). No choice of development. Adobe RGB needs a colour-managed viewer or conversion.	Needs the T0_Raw IIQ files and the Phase One SDK runtime (consent to confirm). Slower; needs the SDK/GPU. Not bit-repeatable with NR on (±1-2/255 on ~2% of pixels). Colours differ from the JPEGs (less green), so indices are not interchangeable. Strong sharpening may halo at canopy stage (not yet tested).

1. Geometry is unchanged from 3.1: the plot is projected into every aligned camera, and the photo holding the whole plot comes first (most central of those); if none holds it all, the photo holding the largest part. The pixel box (and mask) is computed in full-size frame pixels; projects aligned on smaller images are scaled exactly to the IIQ frame.
2. p1tool asks the SDK to develop only that box (SetCrop) at full scale. Developing a box gives the same pixels as developing the whole frame and cutting, at a fraction of the time.
3. The box is masked / georeferenced / written exactly as a JPEG-sourced crop; EXIF gets PIXEL_SOURCE and IIQ_SETTINGS tags, the manifest a pixel_source column, and the settings file is copied next to the output (IIQ_settings_used.json, IIQ_run.json).
4. The IIQ files are only read: an output folder inside the IIQ folder is refused, and every IIQ used is re-checked (size and time) after the run.

APPN HR setting (config_hr)	Value	Effect
Luminance noise reduction	0.8 (single-pixel NR on)	removes sensor grain in brightness
Colour noise reduction	0.8	removes colour speckle
Sharpening	amount 3.1, radius 1.2, threshold 0.2	strong unsharp mask: crisper edges; may halo at canopy stage
Highlight / shadow recovery	on	keeps detail in bright stubble and dark shadows
White balance	as shot	same as the camera's setting
Geometric (lens) correction	on (method 1)	Phase One lens profile, as in the JPEGs
Output	8-bit RGB, sRGB, scale 1.0	full resolution; no ICC profile embedded (sRGB assumed)

Two other presets ship with the app: APPN template (their weaker template) and Neutral (all processing off; least-processed pixels, for spectral measurements, but noisy). The SDK's noise reduction is not bit-repeatable: re-developing the same box can differ by 1-2/255 on ~2% of pixels; neutral settings are exactly repeatable.

4. Experiments and results

E1. Raw crops against the original photos



Figure 3. *Plot 3003: raw and rectified crops; zoom of the original JPEG and the raw crop (difference 0).*

Measure	Result
Plots	36 / 36
Pixels compared (inside plot + margin)	686,638,625
Identical pixels	686,638,625 (100%)
Largest difference	0
Colour profile	Adobe RGB ICC embedded, byte-identical to the source
File	8-bit RGB GeoTIFF, LZW (lossless), EXIF copied

Conclusion: the masked raw crops are the original images, cut. The comparison is against the T1_Proc JPEGs, which are themselves derived from the T0_Raw IIQ files; nothing in PlotExtractor changes them further.

E2. Rectified crops

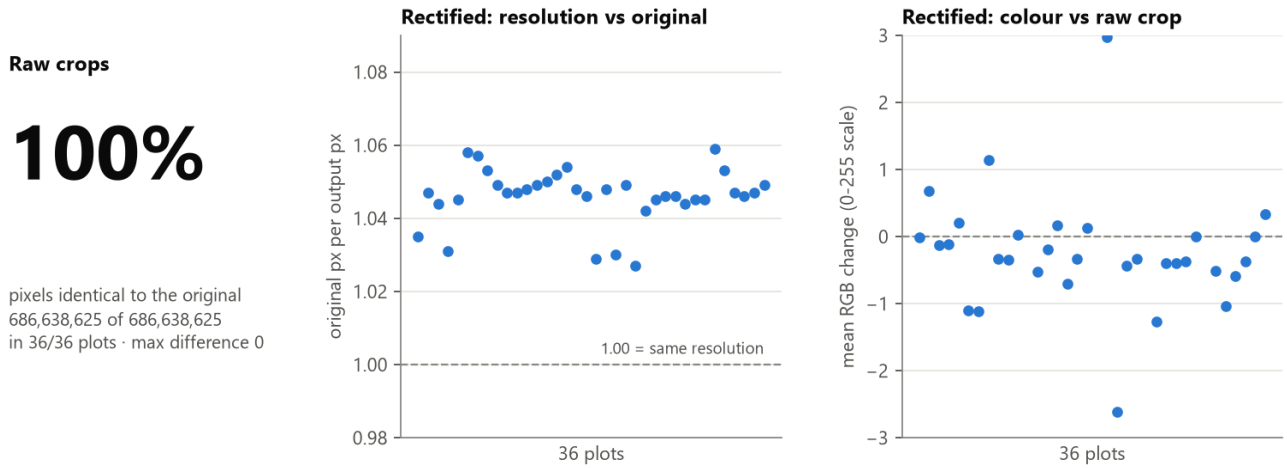


Figure 4. *Rectified crops: 1.03-1.06 original pixels per output pixel; colour within a few levels of the raw crop.*

17 plots came from one photo and 19 needed two. Resolution is 3-6% lower than native; the mean RGB change against the raw crop is small (median −0.2 to −0.3 of 255). Per-pixel PSNR against a re-warped original is low (~16 dB) because the check itself resamples again and any sub-pixel shift on soil texture counts as error; it is not a quality score. Rectified crops look the same as the originals but are not bit-exact.

E3. Grid offset, margin and sown rows

First runs found 4 rows instead of 5 on part of the plots, and on some the middle band shifted by a row. Two causes: the grid sits about 9 cm off the rows across the plot, so the plot edge cut through the outer row and the 0.3-pitch edge rule dropped it; and a comb tooth beyond the data edge got the highest contrast (an edge effect of the detrended profile).

Fixes: teeth outside the image data are excluded; sown rows per plot can be given (5); the plots were re-extracted with a 0.10 m margin (and rectified crops now honour the margin). Result: all 36 plots, raw and rectified, 5 rows found, middle rows 2-4 of 5.

E4. Row detection over 36 plots

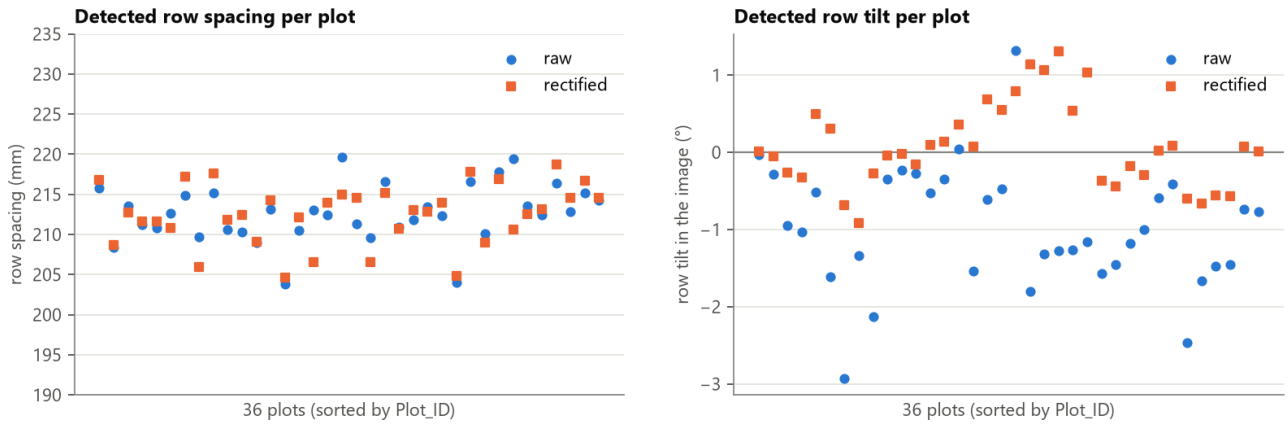


Figure 5. *Detected row spacing and tilt per plot, raw and rectified crops.*

	Raw	Rectified
Plots with 5 rows	36/36	36/36

	Raw	Rectified
Row spacing	204-220 mm, mean 212.6	205-219 mm, mean 212.5
Largest |tilt|	2.93°	1.31°
Middle rows used	2-4 of 5 (all)	2-4 of 5 (all)

The spacing agrees between the two sets and with the expected ~21 cm seeder spacing. Raw crops show more tilt because the plots are not straightened; the tilt estimator follows it. An earlier global angle search picked 3.2° on a plot whose rows were at ~0.7° and was replaced by the segment-phase line fit (3.2, step 3).

E5. Tiles away from the bare plot ends

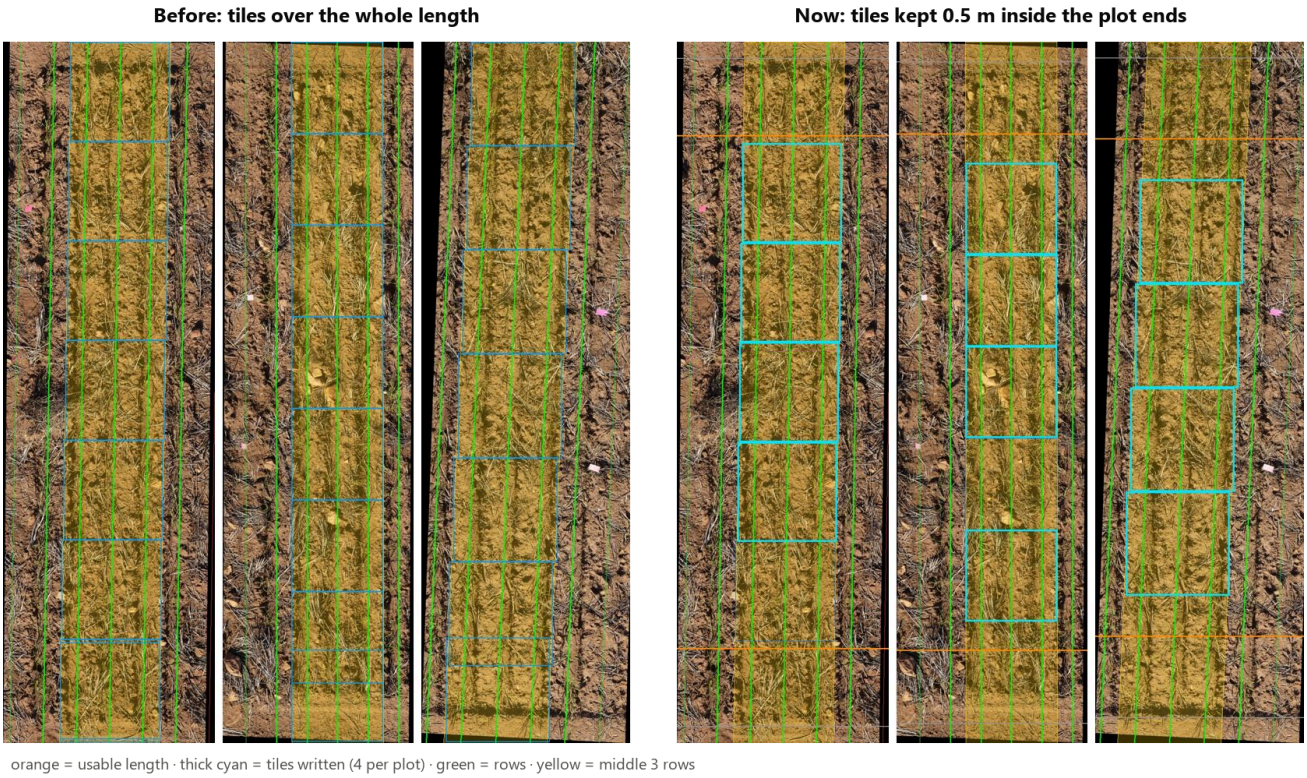


Figure 6. *Before: tiles over the whole length including the bare ends. After: tiles 0.5 m inside the plot ends, centred.*

Measure	Raw	Rectified
Tiles (4 per plot)	144	144
Mean usable length	3.46 m	3.51 m
Tiles outside the usable length	0	0
Tiles not pixel-identical to the crop	0	0
Nearest tile to a crop end	1111 px (~0.6 m)	991 px
Plots with fewer than 4 tiles	0	0

At this seedling stage the planted-length trim did not move the limits (emergence is even along the plots), so the plot ends minus 0.5 m decided the usable length on every plot.

E6. Why 4 plots are not whole in one photo

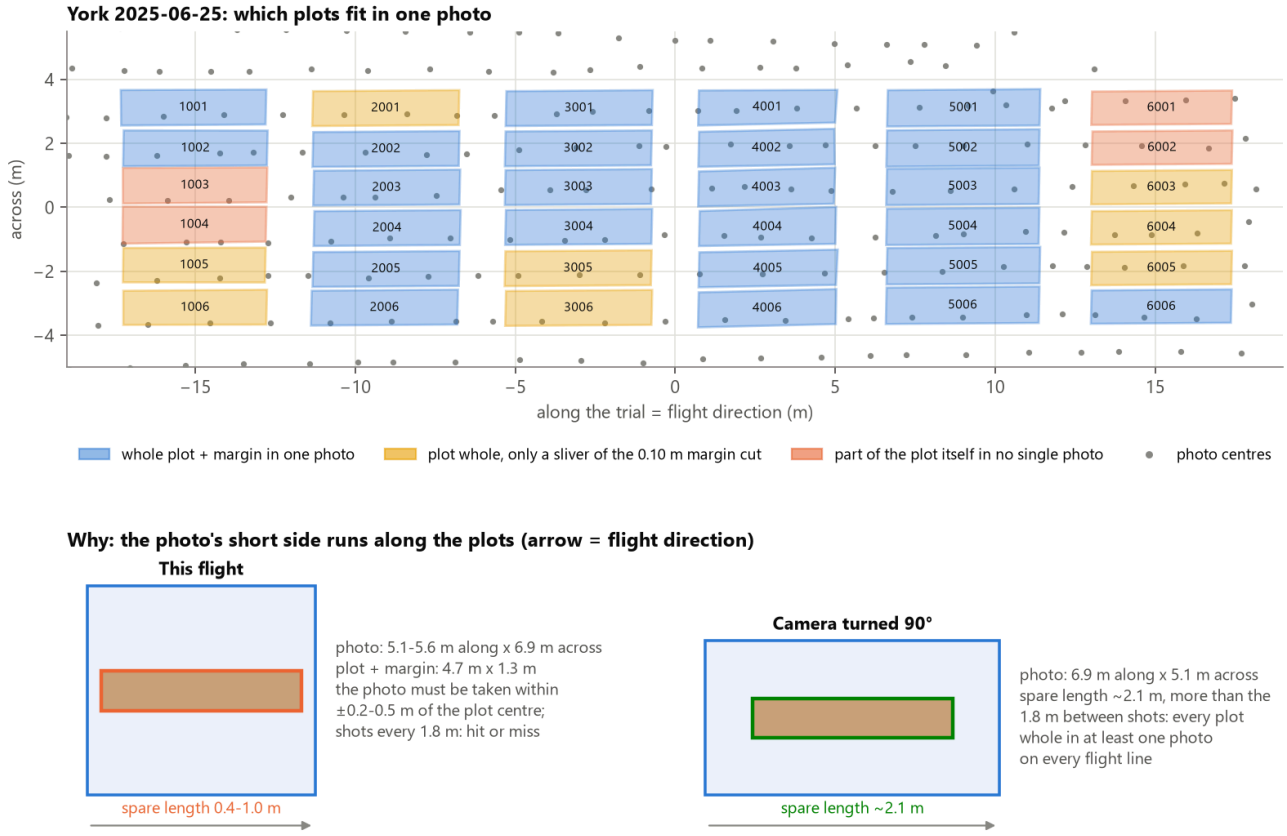


Figure 7. *Top: plots coloured by fit, with photo centres. Bottom: footprint of a photo along the plot, as flown and with the camera turned 90°.*

Plot	Best single photo	Plot covered	Reason
6001	P0019784	96.1%	range at the turn-around end; nearest flight lines start 2-3 m short
1003	P0019809	98.0%	ground ~1 m higher: photo covers 5.1 m along, 0.4 m spare
1004	P0019809	99.3%	as 1003
6002	P0019790	99.6%	as 6001

The camera's short side (9564 px) ran along the plots, i.e. along the flight lines. That side covers 5.1-5.6 m of ground (0.54-0.59 mm/px), the plot + margin is 4.7 m, so a photo must be taken within ±0.2-0.5 m of the plot centre. Shots are 1.8 m apart and lines 1.3 m apart, giving each plot about 4 chances. Range 1 (DEM 260.2 m vs 259.2 m at range 6) is closer to the camera, so its footprint is smaller and its spare length only ~0.4 m. With the margin included, 12 plots do not fit wholly; for 8 of them only a sliver of the margin is cut.

With the camera turned 90°, the long side (12768 px ≈ 6.9 m) runs along the plots, leaving ~2.1 m spare, more than the 1.8 m between shots, so every plot is whole in at least one photo on every line.

E7. IIQ-Direct (APPN HR) against the JPEG-based crops

Two extractions of the same 36 plots (26 Oct 2026): 'our tool' = Extract plots tab, T1_Proc JPEGs, Native box, 0.20 m margin; 'IIQ-Direct' = IIQ plot extraction tab, T0_Raw IIQ files, APPN HR settings, Masked to plot, 0.10 m margin. The two runs named the plots from different shapefile fields (1001-6006 vs 1-36); they were paired by their identical candidate photos and covered areas. Where both used the same photo, the two images were compared on the same photo pixels inside the plot (JPEG converted from its Adobe RGB profile to sRGB first). Noise = Immerkær estimate (median absolute response of a 3x3 Laplacian-difference kernel); colour noise = the same on the two chroma channels; fine detail = standard deviation of the image minus its Gaussian blur (σ 1.5 px).



Figure 8. *Same ground, same photo, 2x zoom: JPEG-based crop (left) and IIQ-Direct crop with APPN HR settings (right). Top: plot 6006 (IIQ plot 1); bottom: plot 2004 (IIQ plot 27).*

Check	Our tool (T1_Proc JPEG)	IIQ-Direct (APPN HR)
Plots	36 / 36	36 / 36 (+ a stray plot_unnamed.tif from an earlier run)
Plots not whole in any photo	4 (96.1-99.6%)	the same 4
Fidelity to own source	36/36 identical to the JPEG box (max diff 0)	6 re-developed: max diff 1/255 on 0.02% of pixels
Same photo as the other run	29 / 36	29 / 36
Alignment (29 plots)	—	0.01-0.02 px shift, correlation 0.984-0.993
Brightness noise, median (range)	3.55 (3.34-3.76)	3.13 (2.72-3.76); lower on 28/29
Colour noise	0.84	0.84 (equal)
Fine detail	19.5 (17.5-21.2)	24.0 (20.9-26.3); higher on 29/29
Detail / noise	5.4 (5.0-5.7)	7.4 (6.8-8.1); higher on 29/29
Mean colour, IIQ minus JPEG (sRGB)	—	R −0.4, G −1.6, B −2.1 levels; ΔE mean 5.2, 95th pct 11.6
Green cover (ExG > 0.1), median	3.0%	2.3% (lower on 29/29; ranking r = 0.93, mean ExG r = 0.996)
Clipped highlights	1.4%	1.9%
Embedded colour profile / tags	Adobe RGB ICC; no EXIF in this run	none (sRGB); camera EXIF + source/settings tags

The 7 plots on different photos (IIQ 12, 16, 17, 18, 21, 24, 28) had several photos holding 100% of the plot, all near-nadir (anisotropy ≤ 1.023) at 0.55-0.60 mm/px; the IIQ tab ranks whole-plot photos by centrality, the earlier run by the tiered order. Either choice is sound.

Conclusion: both sets are complete, correctly placed and true to their sources. IIQ-Direct with the APPN HR settings gives the higher-quality plot images: less noise, more detail, lossless from the raw file, and no sharpening halos visible at this seedling stage. Its colour rendering differs from Phase One's JPEG, which shifts vegetation indices (lower ExG and green cover) while keeping the plots in the same order, so one source should be used per trial and season. This agrees with the Northam 2026 test (plot 1001: detail/noise 2.42 IIQ HR vs 2.15 JPEG). Results: F:\Ibrahim's Workspace2\Plot extraction processing\2025_Calibrationtrial_I_SIF_DPIRD\comparison (REPORT.txt, iiq_vs_ours_per_plot.csv, source_fidelity.csv).

5. Limitations

- Row detection needs visible rows. At emergence the Otsu split marks many soil/stubble pixels as greener, but the rows still show as periodic peaks; with no periodic signal a plot is skipped and reported in rows_report.csv.
- The sown-rows setting is the most reliable way to pick the right middle rows when the grid clips an outer row.
- Rectified images cannot be bit-exact; when original pixels matter use the masked raw crops.
- Plots not whole in any single photo stay cut in the raw set; only the rectified set fills them (from a second photo).
- Tested on one trial and one date (5 rows, seedling stage). Other row counts (e.g. 7) and later growth stages should be spot-checked with the QA pictures.
- IIQ-Direct: APPN's sharpening (3.1) is strong; check leaf edges for halos on a canopy-stage flight before relying on it. Its colours differ from the JPEGs, so vegetation indices are not interchangeable between the two sources. It needs the Phase One SDK runtime (shipped with the app; confirm that the DPIRD Phase One consent covers PlotExtractor) and is slower than cutting JPEGs.

6. Recommendations

- Extract with Masked to plot, GeoTIFF, sRGB off and a 0.10 m margin for original-quality plot images.
- For the highest image quality, use the IIQ plot extraction tab (APPN HR settings) on the T0_Raw IIQ files; use the Neutral preset when least-processed pixels matter more than looks. Keep one pixel source per trial and season.
- Tile from the raw crops with middle 3 rows, sown rows set, tiles kept 0.5 m from the ends, PNG.
- Check _rows_qa for a few plots per run and the raw_covered_pct column of the manifest.
- Plan future Phase One flights with the camera's long side along the plots (or 2-3 m higher), and extend flight lines past the trial ends.

7. Files and reproducibility

What	Where
Application (source + exe)	E:\Software\PlotExtractor-2026-10-24 (dist\PlotExtractor\PlotExtractor.exe); previous version E:\Software\PlotExtractor-2026-09-01 unchanged
New / changed code	row_detect.py (new), tiler.py (middle rows, usable length, tiles per plot, QA), plot_extractor.py (manifest box, margin for rectified, mask to plot + margin), PlotExtractor.spec
York outputs	…\PlotExtractor_2026-10-24_York_2025-06-25\raw, \rectified (RawCrops, manifest.csv, candidates.csv, run_log.txt, Tiles_middle3rows\)
Fidelity results	…\fidelity\ (FIDELITY_SUMMARY.txt, raw_vs_original.csv, rectified_vs_original.csv, fidelity_check.py)
IIQ-Direct vs JPEG (E7)	F:\Ibrahim's Workspace2\Plot extraction processing\2025_Calibrationtrial_I_SIF_DPIRD\ (our tool\, IIQ-Direct Processing\, comparison\: REPORT.txt, compare_crops.py, check_sources.py, CSVs, zoom picture)
IIQ code	iiq.py (tabs, Developer, georeferencing), iiq_tools\src\p1tool.cpp (adapted from APPN-UQ HR_clip, MIT), iiq_tools\presets\, THIRD_PARTY_NOTICES.md
Docs	docs\PlotExtractor_User_Guide (.docx/.pdf), PlotExtractor_Technical_Report (.docx/.pdf), PlotExtractor_Demo.mp4, figures\